MIL-S-20441C (PA)
28 January 1974
SUPERSEDING
MIL-S-20441B (MU)
25 July 1967

MILITARY SPECIFICATION

SIMULATOR, FLASH ARTILLERY, M110 PARTS, AND LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts manufacture, and loading, assembling and packing for one type of simulator, designated as Simulator, Flash Artillery, M110.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-A-2550	- Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC: 1370

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MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY PICATINNY ARSENAL

PRODUCT AND PACKAGING DRAWINGS

8848614 - Simulator, Flash Artillery, M110
2128171 - Pack, Shipping and Storage (For 8 Simulator, Flash Artillery, M110 with 8 Squib, Electric)
3139673 - Package, Shipping and Storage (For 4 Simulator, Flash Artillery, M110 with 4 Squib, Electric)
8880486 - Box, Packing, Ammunition, For Simulators, Flash Artillery, M110 and Squibs, Electric 593
8880487 - Container, Ammunition, Fiber, M242, for Simulator, Flash Artillery, M110

INSPECTION EQUIPMENT DRAWINGS

9221518 - Tube, Firing for M110 Simulator
9221615 - Test and Evaluation Procedure for M110 Simulator

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

*2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transporation 0-190

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(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The Simulator Assembly shall comply with all requirements specified on Drawing (dwg.) 8848614, and all associated drawings, and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Charge composition.-The moisture content of the charge composition at the loading station at time of loading shall not exceed 0.15 percent.

3.4 Functioning - Proving Ground.-Following the addition of gasoline the simulator shall comply with the following requirements.

3.4.1 The simulator shall not fail to function.

3.4.2 The simulator shall function with a bright yellow flash.

3.4.3 The size of flash shall be not less than nine (9) feet long and not less than six (6) feet wide.

3.4.4 The sound intensity at twenty-five (25) plus or minus 0.5 feet from the implacement point of the simulator shall be not less than 120 decibels (DB).

3.5 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease and other defects and foreign matter which would affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

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3.6 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements when such material is purchased by the contractor or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First article inspection.

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 complete sets of parts and sub-assemblies (i.e. 25 each of every component and sub-assembly, including at least three (3) items from each cavity used to produce plastic parts and be so identified as to permit tracing each item to the cavity in which it was molded) and 50 loaded assemblies, completely packaged and sealed which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 First article inspection provisions

4.2.2.1 Parts and sub-assemblies (inert).-The 25 sets of parts and sub assemblies (see 4.2.1) shall be submitted to a Government approved laboratory designated by the Contracting Officer. They shall be subjected to any or all of the applicable examinations and tests specified in 4.3.2, 4.3.3 and 4.3.5 of this specification and any or all of the requirements of the applicable drawings. If any part or subassembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon encountering any failure of an item to comply with any of the specified requirements.

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4.2.2.2 Loaded simulator assemblies.-The 50 loaded simulator assemblies (see 4.2.1) shall be submitted to a Government proving ground designated by the contracting officer. They shall be subjected to the functioning test in accordance with the procedure specified in 4.4.2. The first article sample shall be rejected if three (3) or more simulators fail to comply with the requirements of 3.4. The Government reserves the right to terminate its inspection upon encountering, any failure of an item to comply with any of the specified requirements.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of simulators shall contain:

a. Parts of one kind from one supplier
b. Composition ingredients from not more than one lot from one supplier
c. Composition from one batch or blending of batches that result in homogeneity of the lot
d. Electric squibs of one designation from one supplier

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and

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major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Simulator parts assembly (see dwg. 88480609 covering a detail of dwg. 8848614).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Any component missing	Visual	07002
102.	Cement composition missing from joints or inadequate	Visual	07003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	07004

4.3.2.2 Assembly prior to loading (see dwg. 8848614).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Evidence of poor workmanship (see 3.5)	Visual	08001
Minor:	None defined		

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4.3.2.3 Loading assembly, prior to assembling adapter holder (see dwg. 8848614).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Simulator charge composition missing	Visual	09001
102.	Simulator charge composition deposited on threads of the adapter holder seat	Visual	09002
Minor:	None defined		

4.3.2.4 Loading Assembly (see dwg. 8848614).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Total length, max.	Gage	10001
102.	Cement composition missing on adapter holder threads or inadequate	Visual	10002
103.	Any component missing	Visual	10003
Minor:	AQL 1.00 percent		
201.	Marking misleading or unidentifiable	Visual	10004
202.	Evidence of poor workmanship (see 3.5)	Visual	10005

4.3.2.5 Container, prior to sealing (see dwg. 8880487).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Simulator inverted	Visual	11001
102.	Packing components missing	Visual	11002
Minor:	None defined		

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4.3.2.6 Container, sealed (see dwg. 8880487).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Container damaged	Visual	12001
102.	Contents loose	Manual	12002
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	12003

4.3.2.7 Sealed package (see dwg. 8880486).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Seal improperly located or incomplete	Visual	13001
102.	Bag punctured, torn or cut	Visual	13002
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	13003

4.3.2.8 Wood packing box (see dwg. 8880486).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Contents exposed due to damaged box	Visual	14001
102.	Hardware or strapping missing, broken or loose	Visual/Manual	14002
103.	Box broken, board loose or protruding nails	Visual	14003

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Minor:	AQL 1.50 percent		
201.	Contents loose	Manual	14004
202.	Hardware or strapping improperly assembled	Visual/Manual	14005
203.	Marking misleading or unidentifiable	Visual	14006

*4.3.2.9 Package, Shipping and Storage prior to closing (see dwg. 3139673)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Any packing component missing or misassembled	Visual	16001
Minor:	AQL 2.50 percent		
201.	Number of dessicant units incorrect	Visual	16002
202.	Dessicant bags not properly secured	Visual	16003
203.	Indicator card missing	Visual	16004
204.	Number of inner packages incorrect	Visual	16005
205.	Evidence of poor workmanship (See 3.5)	Visual	16006

*4.3.2.10 Package, Shipping and Storage (See 3139673)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	CFR markings missing or unidentifiable	Visual	18001
Major:	AQL 0.40 percent		
101.	Tray interferes with seal of lid	Visual/Manual	18002
Minor:	AQL 1.00 percent		
201.	Descriptive marking missing misleading or unidentifiable	Visual	18003
202.	Evidence of poor workmanship (see 3.5)	Visual	18004

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*4.3.2.11 Pack, Shipping and Storage, prior to closing (see dwg. 2128171).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Package missing, or improperly positioned	Visual	19001
Minor:	None defined		

*4.3.2.12 Pack, Shipping and Storage (see dwg. 2128171).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	CFR markings missing or unidentifiable	Visual	20001
Major:	AQL 0.40 percent		
101.	Wire binding not properly secured	Visual	20002
102.	Car seal missing	Visual	20003
Minor:	AQL 1.00 percent		
201.	Descriptive marking missing misleading or unidentifiable	Visual	20004
202.	Evidence of poor workmanship (see 3.5)	Visual	20005

4.3.3 Production test.-The following test shall be performed using Standard MIL-STD-105 for sampling plan requirement unless otherwise specified.

4.3.3.1 Moisture content (see Table I) - Major defect

TABLE I

Material	Code No.
Charge composition (see 3.3.1)	15001

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The contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and subject to test, one sample of each material from each eight hours production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with a material containing excess moisture, the remaining unloaded material and the loaded components shall be rejected. Test procedures shall be as specified in 4.4.1.

4.3.3.2 Functioning.-The simulator assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table II, when tested as specified in 4.4.2.

TABLE II

Defect	Classification	Code No.
Simulator fails to function (see 3.4.1)	Major	17001
Color of flash (see 3.4.2)	Major	17002
Volume of flash (see 3.4.3)	Major	17003
Sound intensity (see 3.4.4)	Major	17004

4.3.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, fifty (50) simulator assemblies shall be selected for this test. The lot shall be rejected if, during the test, three (3) or more simulator assemblies exhibit any of the major defects listed in Table II.

4.3.3.2.2 After three (3) consecutive lots have complied with the criteria of 4.3.3.2.1, thirty-two (32) simulator assemblies shall be selected from each lot for test. The lot shall be rejected if, during the test, two (2) or more simulator assemblies exhibit any of the major defects listed in Table II.

4.3.3.3 Heat seal test of sealed bag (see dwg. 8880486) - Major defect - Code No. 13004.-Sample size and acceptance criteria shall be as specified in MIL-P-116. Testing shall be as specified in 4.4.3 (non-destructive test).

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4.3.3.4 Leakage test (see dwg. 8848609) - Major defect - Code No. 07005.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.4. Any assembly that fails to comply with the requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.3.4 Inspection Equipment.-Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approval.

*4.3.5 Dimensional control of molded parts.-In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 1) a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.3.5.1 through 4.3.5.4 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL Of 0.40 percent for each major defect and an AQL Of 0.65 percent for minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantitate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, Picatinny Arsenal, Dover, New Jersey 07801, Attn: SARPA-QA-A-M for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

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NOTE 1.-In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey 07801 Attn: SARPA-QA-A-M. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time shall be established and approved.

*4.3.5.1 Plug, filling (see dwg. 8848605 covering a detail of dwg. 8848614).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Pitch diameter of thread	Gage	01001
102.	Major diameter of thread	Gage	01002
103.	Concentricity of outside diameter with major diameter of thread	Gage	01008
Minor:			
201.	Total length	Gage	01003
202.	Length under head	Gage	01004
203.	Large outside diameter, min.	Gage	01005
204.	Width of slot	Gage	01009
205.	Depth of slot	Gage	01010

*4.3.5.2 Container, inner (see dwg. 8848607 covering a detail of 8848614).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Total length, min.	Gage	02001
102.	Wall thickness, min.	Gage	02002
103.	Thickness thru bottom	Gage	02003

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104.	Inside diameter at bottom	Gage	02013
105.	Depth of thread cavity	Gage	02004
106.	Pitch diameter of thread	Gage	02005
107.	Minor diameter of thread, max.	Gage	02006
Minor:			
201.	Outside diameter	Gage	02008
202.	Diameter of flange	Gage	02009
203.	Diameter under flange	Gage	02010
204.	Length of flange	Gage	02011
205.	Inside diameter at open end	Gage	02014
206.	Depth of thread cavity counterbore	Gage	02015

*4.3.5.3 Holder, adapter (see dwg. 8848606 covering a detail of dwg. 8848614).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Total length, min.	Gage	03001
102.	Pitch diameter of thread, min.	Gage	03002
103.	Major diameter of thread, min.	Gage	03003
104.	Wall thickness, min.	Gage	03004
105.	Thickness thru bottom	Gage	03005
106.	Depth of web	Gage	03006
107.	Inside diameter at bottom of cavity, min.	Gage	03007
108.	Inside diameter at top of cavity, max.	Gage	03008
109.	Outside diameter of web	Gage	03009
110.	Inside diameter of web	Gage	03010
Minor:			
201.	Length of flange	Gage	03012
202.	Length of thread, pilot, max.	Gage	03013
203.	Width of ribs	Gage	03014
204.	Outside diameter under flange, max.	Gage	03015
205.	Countersink missing	Visual	03016

*4.3.5.4 Container, outer (see dwg. 8848608 covering a detail of dwg. 8848614).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Total length, min.	Gage	04001
102.	Wall thickness, min.	Gage	04002
103.	Diameter of thread counterbore	Gage	04003
104.	Depth of thread counterbore	Gage	04004
105.	Pitch diameter of thread	Gage	04005
106.	Minor diameter of thread, max.	Gage	04006
107.	Concentricity of thread counterbore with minor diameter of thread	Gage	04015
Minor:			
201.	Large outside diameter	Gage	04007
202.	Small outside diameter	Gage	04008
203.	Large inside diameter	Gage	04009
204.	Small inside diameter	Gage	04010
205.	Diameter of boss at threaded end	Gage	04011
206.	Length of flange	Gage	04012
207.	Outside diameter of flange	Gage	04016
208.	Marking misleading or unidentifiable	Visual	04013

4.4 Test Methods and Procedures

4.4.1 Moisture content

4.4.1.1 Charge composition.-Approximately five grams (gms.) shall be accurately weighed and placed in a tared weighing dish. The dish, cover and contents shall be weighed and then placed in an oven and dried at 100 plus or minus 2 degrees Centigrade (C.) for 4 hours. The dish and contents shall be removed from the oven, cooled in a desiccator and weighed. The loss in weight shall be calculated as percent moisture in the sample.

4.4.2 Functioning (Proving Ground).-The sample simulators selected for test shall be prepared by removing the red filling plug and inserting 70 ml. of 90 octane gasoline. Replace the red filling plug and screw it down tightly. The electric squib, is then inserted in the central hole of the adapter. The

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lead wires of the electric squib shall be drawn across the base and along the body of the simulator toward the red filling plug end, where they are to be secured to the simulator body by an adhesive tape. The simulator shall be placed, adapter end first, in a fixture and fired with a current of 0.50 plus or minus 0.05 amperes applied to the electric squib. Equipment and procedure shall be as specified in dwgs. 9221518 and 9221615. All observers shall be behind cover at least 50 yards away from the firing tube. The angle of elevation of the firing tube shall be approximately 45 degrees. All combustible material, such as dry grass, is to be removed for a radius of 2 yards from the firing fixture.

4.4.2.1 Test validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.4.3 Heat seal test of sealed bag.-This test shall be conducted in accordance with the procedures specified in Specification MIL-P-116.

4.4.4 Leakage.-Each assembly shall be placed in a tank of water and submerged to a minimum depth of 1 inch beyond the cemented joint. The immersed assembly shall withstand an air pressure of 5 pounds per square inch min., without evidence of leakage. The equipment necessary to conduct this test shall be in accordance with 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The simulators shall be preserved and packaged in accordance with dwg. 8880486.

*5.1.2 Level A.-For Marine Corps use, preservation and packaging shall be in accordance with dwg. 3139673.

5.2 Packing

5.2.1 Level A.-The cartons shall be packed in accordance with dwg. 8880487.

*5.2.2 Level A.-For Marine Corps use, packing shall be in accordance with dwg. 2128171.

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5.3 Marking

5.3.1 Marking.-The cartons shall be marked as specified in dwgs. 8880486, and 8880487.

*5.3.2 For Marine Corps use, marking shall be as specified in dwgs. 3139673 and 2128171.

6. NOTES

*6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

d. For Marine Corps packing, packaging and marking see 5.1.2, 5.2.2. and 5.3.2.

*6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3.)

*6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

*6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

*6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirement Lists (DD Form 1423).

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

*6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

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*6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*6.5 Proving ground functioning test summary

Test	Production lots			Requirements
	First Article	First three Consecutive Acceptable	Remaining	
Function	50	50	32	3.4 (see Table II)

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1370-A-024

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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